Forward Collision Warning System Considering Both Time-to-Collision and Safety Braking Distance

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Abstract- A novel algorithm considering both time- to-collision (TTC) and safety braking distance for forward collision warning system is presented for to alert and to assist driver keeping safety braking distance to avoid the collision accident in highway driving. We calculate the safety braking distance and TCC based on the most important parameters which are the distance between the driving car and the vehicle (obstacle) ahead, the variable of the distance between the driving car and the vehicle (obstacle) ahead, vehicle weight, vehicle speed, slope of road, condition of road surface and the age of driver. The system compares the distance of vehicle (obstacle) ahead and safety braking distance and to determine the moving vehicle's safety braking distance is enough or not. The reaction time of driver and pressure buildup time of braking system are all taken into account. The useful alert messages can serve as a safety assistance system for safer driving in highway.

Keyword–time to collision, vehicle safety braking distance

1. INTRODUCTION

In order to improve the drive safety, many researchers have devoted efforts on automobile anti-collision technology in recent years [1-16]. The statistics show that around 21% of collision accidents are forward collision accidents. To avoid the forward collision accidents is very important and the modern cars are equipped with all kinds of measuring and alert system in order to assist driver in safety driving status. The passive safety driving assistance system played a main role in the past and the active safety driving assistance systems so-called advanced driver assistance systems (ADAS) will be the trend for future vehicle development.

Masumi Nakaoka1 et al. [3] focused on a forward collision warning algorithm based on road friction coefficient and driver characteristics. Jing-Fu Liu et al. [4] developed an advanced driver assistance system with lane departure warning and forward collision warning functions. The driving simulator experiments with 65 participants and investigated drivers acceptance of multiple warnings (visual and auditory displays) including forward collision warning system and lateral collision warning system while driving on Metropolitan Expressway were described in [7]. Stefan K. Gehrig and Fridtjof J. Stein [13] introduced a planning and decision component to generalize vehicle following to situations with nonautomated interfering vehicles in mixed traffic.

The salient features of this paper are summarized as follows:

(1) A useful structure of the vehicle safety distance warning system considering both TTC and safety braking distance for to assist the driver driving in safety condition is presented.

(2) The presented algorithm can calculate the real time TTC and safety braking distance for moving vehicle considering the real time of driving conditions.

(3) The presented algorithm can inform driver the messages of TTC and braking distance between the moving vehicles. Four level warning signals for to assist driver to drive in safety are presented in this paper.

2. STRUCTURE OF FORWARD COLLISION WARNING SYSTEM

A vehicle safety distance warning system uses ultrared ray reader and transmitter to detect the distance between driving vehicle and car (obstacle) ahead and deliver necessary information to the calculating unit is shown in figure 1[14]. The proposed algorithm will utilize the vehicle speed, vehicle weight, weather condition, slope of the road, road surface, age of driver, and etc. to figure out the minimum safety braking distance ($D_{s,min}$) and maximum safety braking distance ($D_{s,max}$) respectively. The TTC will take into consideration for to inform the driver the driving condition, safety or dangerous. There are four level warning signals for to assist driver to drive in safety.

3. SAFETY BRAKING DISTANCE

The braking distance for the moving vehicle from in speed V to stop completely in speed zero is expressed in equation (1) [17,18].

$$D_b(x)=\frac{\gamma W}{2gC_{ae}}\ln\left(1+\frac{C_{ae}V^2}{\eta_b(\mu+f_r)W\cos\theta_s \pm W\sin\theta_s}\right) \qquad (1)$$

where,

$D_b(x)$: braking distance of moving vehicle to speed zero.
γ : equivalent mass factor.
W : vehicle weight.
g : gravity speed.
C_{ae} =(ρ*A_f*C_d)/2
ρ :mass density of the air.

A_f :characteristic area of the vehicle.
C_d :coefficient of aerodynamic resistance.
V : vehicle speed.
η_b : brake efficiency.
μ : road adhesion coefficient.
f_r : rolling resistance coefficient.
θ_s : angle of the road slope with the horizontal.
$\pm$:positive sign for vehicle moving uphill, negative sign for vehicle moving downhill grade.

The braking distance is the traveling distance for a vehicle from speed V to complete stop after the brake is pushed. The parameter of brake efficiency is depended on braking system, tire condition and so on. Usually, the value of brake efficiency is varies in the range of 80% to 100%. The rolling resistance coefficient will be affected by road condition and the tire rotation speed, and its value varies between 0.007 and 0.015. For the passenger vehicles and light trucks, the equivalent mass factor, γ, is in the range 1.03 to 1.05.

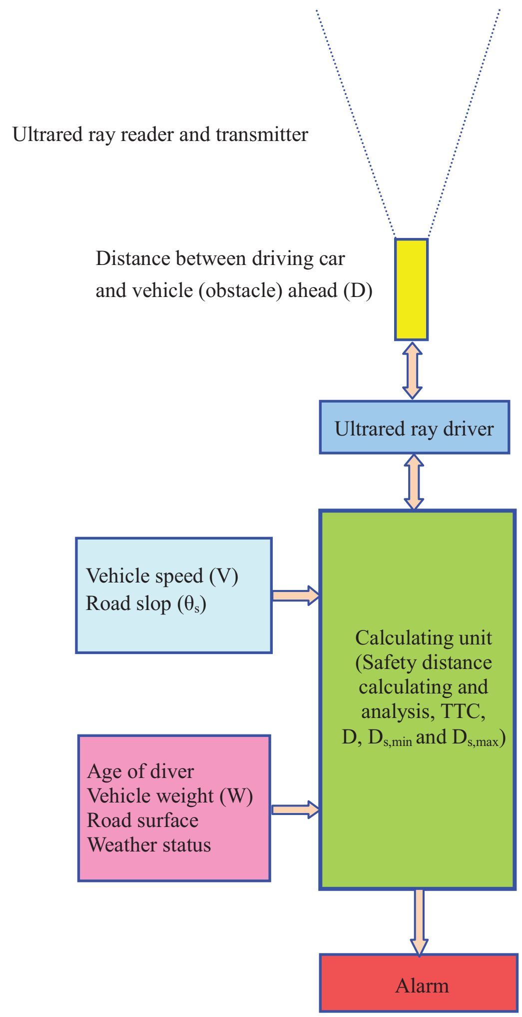

Figure 1 Configuration of vehicle safety distance warning system.

In here, the parameter μ, road adhesion coefficient, is dependent upon the factors of weather, road condition and so on. In addition, the wheel slip ratio, s_r, is defined as equation (2) and the road adhesion coefficient is a nonlinear function of the wheel slip ratio [18].

$$s_r=(V-Rw)/V \tag{2}$$

where R is radius of tire and w is angular speed of tire. The main purpose of ABS is to prevent the wheel slip ratio equal 1 (w=0). When the tire is locked the driving vehicle will lose directional control and stability, and the stopping distance and braking time will be longer than equipped with ABS.

The summary of average values of road adhesion coefficient in different road conditions is list in table 1 [18]. Table 2 summarizes the parameters of the passenger vehicle of Mitsubishi Zinger 2.4 in Taiwan for simulation data.

Table 1 Average values of road adhesion coefficient in different road conditions.

Pavement		With ABS	Setting	Without ABS	Setting
Asphalt	Dry	0.8-0.9	μ_{min}=0.8 μ_{max}=0.9	0.75	μ_{min}=0.75 μ_{max}=0.75
	Wet	0.5-0.7	μ_{min}=0.5 μ_{max}=0.7	0.45-0.6	μ_{min}=0.45 μ_{max}=0.60
Concrete	Dry	0.8-0.9	μ_{min}=0.8 μ_{max}=0.9	0.75	μ_{min}=0.75 μ_{max}=0.75
	Wet	0.8	μ_{min}=0.8 μ_{max}=0.8	0.7	μ_{min}=0.70 μ_{max}=0.70
Snow		0.2	μ_{min}=0.2 μ_{max}=0.2	0.15	μ_{min}=0.15 μ_{max}=0.15
Ice		0.1	μ_{min}=0.1 μ_{max}=0.1	0.07	μ_{min}=0.07 μ_{max}=0.07

Table 2 Simulate parameters setting of Mitsubishi Zinger 2.4.

Parameter	Value range or conditions	Setting for simulation
W, vehicle weight	-	Zinger(1,620 kg) + one person weight (60 kg)= 1,680 kg
g, gravity acceleration	-	9.8(m/s^2)
ρ, air density	in 25°C environment	1.18 kg/m^3
A_f, Projection area	-	High*Wide=1.61m*1.775m
C_d, aerodynamic resistance coefficient	0.15-0.5	0.45
η_b, braking efficiency	0.8-1.0	η_{min}=0.85, η_{max}=0.95
f_r, rolling resistance coefficient	0.012-0.015	0.015
γ, equivalent mass factor	1.03-1.05	1.04

Reaction distance

When driver operates the vehicle, all the data will be transferred through nerve system and processed by human brain. In the emergency condition, after the brain making correct judgment, the command will be transmitted to limbs or other organs. This is a succession of response course, called the "reaction ability". The reaction process, Consciousness→ Judgment→ Action, the time duration that this response course needs, is called reaction time. The distance traveled by vehicle during the reaction time is called the reaction distance. The length of reaction time will be affected by age, distinction, physical stamina, emotion. General speaking, the reaction time can be divided into three components:

(1) *Reflection time:* The time which uses vision, hearing or experience to investigate external conditions, is called reflection or experience time. The reflection time of common people is about 0.44−0.52 second. However, it is different with the driver's emotion and physical stamina. Moreover, road conditions or environment are the factors which must be taken into account too. If road conditions are more complicated, it will lengthen the reflection time. The reflection time varies in the range 0.44 to 0.52 seconds.

(2) *Judgment time:* We use our brain and wisdom to understand, and judge. The time of this process is called judgment time. Judgment time of common people is about 0.15−0.25 second. But it is affected greatly by emotion. When people nervous, or perfectly calm, it will show more differences.

(3) *Action time:* After the judgment to apply an action at the appropriate situation is called action time. The action time of common people is about 0.15−0.4 second.

Finally, the reaction time, t_r, including reflection time, judgment time and action time is vary in range 0.74 to 1.17 seconds depending on age of driver. The summaries of the driver reaction time depending on driver's age are list in table 3. The reaction distance, D_r, is vary in range $t_{r,min}$ V to $t_{r,max}$ V where V is the vehicle speed. The minimum and maximum reaction distance, $D_{r,min}$ and $D_{r,max}$, can be expressed as follow respectively.

$$D_{r,min}= t_{r,min} V \tag{3}$$
$$D_{r,max}= t_{r,max} V \tag{4}$$

Table 3 Reaction time of driver.

Age of driver	Reaction time range	Setting
18-33	0.74-1.10	$t_{r,min}$=0.74 $t_{r,max}$=1.10
34-47	0.75-1.12	$t_{r,min}$=0.75 $t_{r,max}$=1.12
48-57	0.77-1.13	$t_{r,min}$=0.77 $t_{r,max}$=1.13
58-70	0.78-1.17	$t_{r,min}$=0.78 $t_{r,max}$=1.17

Pressure buildup distance

The pressure buildup distance, D_p, is the distance of vehicle travel during the initiation of braking action to the full setup the braking pressure. Usually the pressure buildup time, t_p, varies in the range between 0.3 and 0.75 second for the braking system to build up the braking forces to the between wheels and road. The pressure buildup distance, D_p can be expressed as follow.

$$D_p=t_p V_0-(a_p t_p^2)/2 \tag{5}$$

where V_0 is vehicle speed before braking. The a_p is the average deceleration of the vehicle and it could be neglected because the speed of vehicle approximate the same during the pressure buildup period. So the minimum and maximum pressure buildup distance can be presented as equations (6) and (7).

$$D_{p,min}=t_{p,min} V_0 \tag{6}$$
$$D_{p,max}=t_{p,max} V_0 \tag{7}$$

Where $t_{p,min}$ and $t_{p,max}$ are the minimum and maximum pressure buildup time respectively and its values are 0.3s and 0.75s respectively.

Safety braking distance of the driving vehicle

Finally, the safety braking distance of the driving vehicle , D_s, can be described as follow.

$$D_s=D_b+D_r+D_p \tag{8}$$

According to the varies of the reaction time and the pressure buildup time, The maximum safety braking distance and the minimum safety braking distance can be determined respectively as follow.

$$D_{s,max}=D_b(\mu_{min}, \eta_{min})+ D_{r,max} + D_{p,max} \tag{9}$$
$$D_{s,min}=D_b(\mu_{max}, \eta_{max})+ D_{r,min} + D_{p,min} \tag{10}$$

4. ALGORITHM OF FORWARD COLLISION WARNING

A novel algorithm for forward collision warning system is presented in this section. Figure 2 depicts the presented algorithm for forward collision warning system. Firstly, the parameters such as the vehicle with ABS or not, rolling

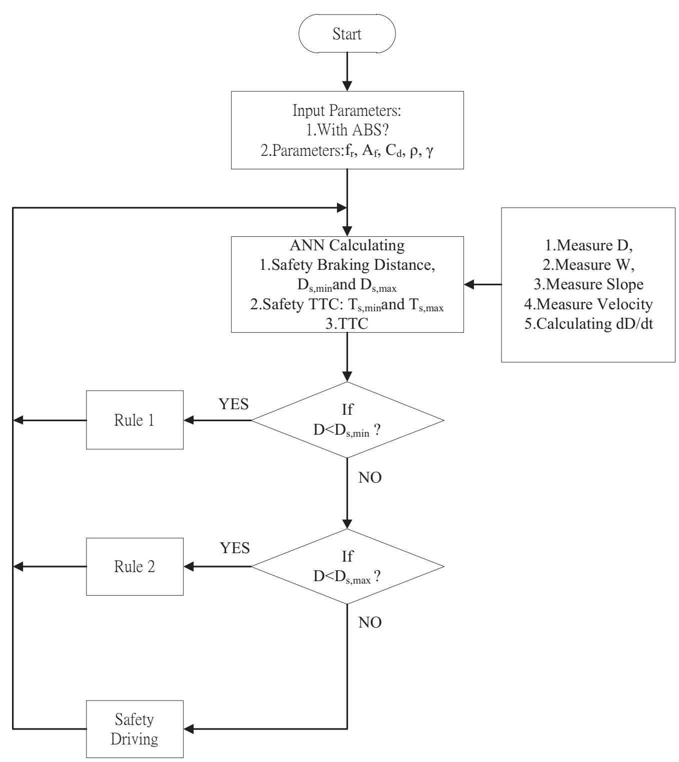

Figure 2 Algorithm of forward collision warning system.

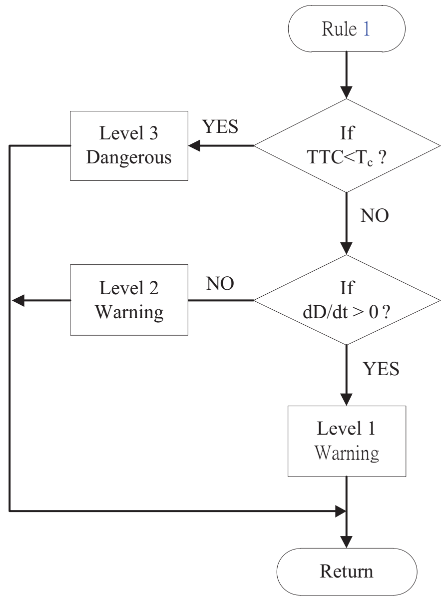

Figure 3 Rule 1algorithm of forward collision warning system.

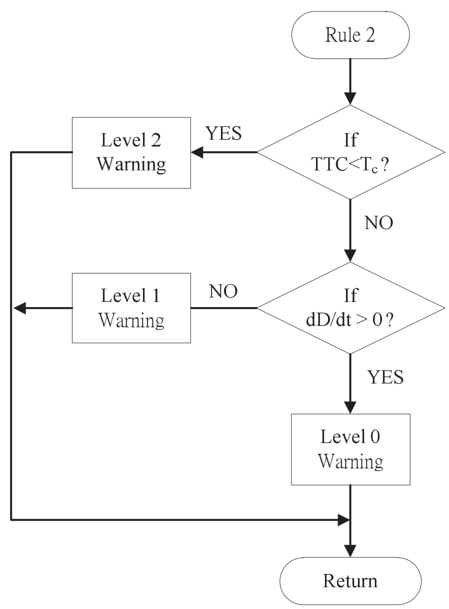

Figure 4 Rule 2 algorithm of forward collision warning system.

resistance coefficient, characteristic area of the vehicle, coefficient of aerodynamic resistance, mass density of the air and equivalent mass factor are necessary to input the system. And then we calculate the maximum and minimum safety braking distance and TTC. At the same time, the critical TTC will be calculated. If the distance between driving cars, D, is shorter than minimum safety braking distance, $D_{s,min}$, then go to rule 1 algorithm. Otherwise, if D is shorter than maximum safety braking distance, $D_{s,max}$, then go to rule 2 algorithm. Otherwise the system will inform that the driving vehicle is under safety driving condition. Figure 3 shows the rule 1 algorithm of forward collision warning system. In the rule 1, the distance between driving cars is shorter than minimum safety braking distance. If TTC is smaller than critical TTC, T_c, then the driving vehicle is under dangerous situation. The driver needs to slow down the velocity to avoid forward collision accident. This is the worst situation mean level 3 warning. In rule 1 algorithm, there are three levels warning signals will be produced, level 3, level 2 and level 1. There are also three levels warning signals will be produced in rule 2 algorithm. Figure 4 presented the algorithm of rule 2.

5. SIMULATION RESULTS

The simulation results are presented in this section. A passenger vehicle with model Mitsubishi Zinger 2.4 in Taiwan for experimental is a weight 1,620 kg vehicle with 2,400 cc engine. Simulate parameters setting of Mitsubishi Zinger 2.4 are listed in table 2. The critical TTC, T_c, is set to be 1.5s[15,16]. The definition of TTC could be expressed as equation (11).

$$TTC=D(t)/(dD(t)/dt) \tag{11}$$

Where D(t) is the distance between driving cars.

A period of 767 seconds start at AM10:21:34 on February 5, 2010 for highway driving is presented for demonstrate the performance of the presented algorithm. Figure 5 shows the RPM of engine of vehicle operating. Velocity of vehicle operating is shown in figure 6. In the beginning 96 seconds, it is an acceleration period. The velocity accelerated from low speed to high speed and keep in high speed for about 600 seconds. Finally, a period about 50 seconds operated for decelerated the velocity until to stop.

Figure 7 described the distance between driving cars during the period. The value beyond 200 meters is not appearing in the figure. The maximum and minimum of safety braking distances of driving vehicles are depicted in figure 8. To compare the distance between driving cars in figure 8, it is obvious to see that the driving cars are not to keep a safety braking distance during at about 140^{th} second to 165^{th} second. The TTC and critical TTC of vehicle operating are shown in figure 9. In the figure, we could find that there are 5 times happening the TTC smaller than critical TTC. Figure 10 presented the critical TTC and warning signal of vehicle operating. The forward collision system informs five type warning signals to driver for warning. The warning levels from level 0 to level 4 are presented in this paper. The level 4 warning signal means the driving car in the most dangerous situation and driver needs to slow down the speed. And the level 0 signal is only warning driver to pay attention in distance between driving cars.

The performance of using presented algorithm in forward collision warning system is guaranteed. The simulation results described the presented method is suitable applying in forward collision warning system for to avoid the forward collision accident.

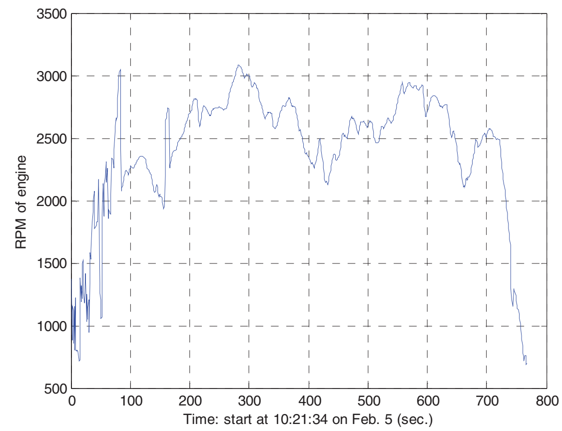

Figure 5 The engine RPM of vehicle operating on February 5, 2010 start at AM10:21:34.

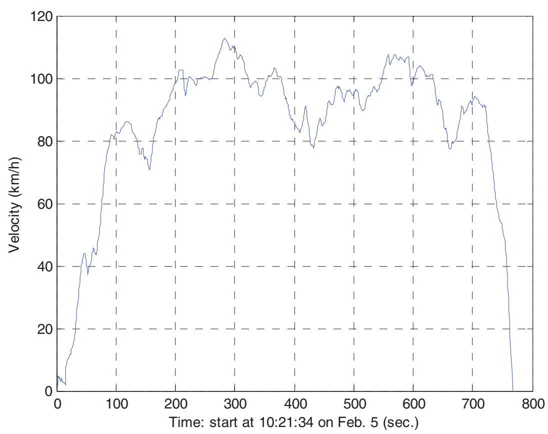

Figure 6 Velocity of vehicle operating on February 5, 2010 start at AM10:21:34.

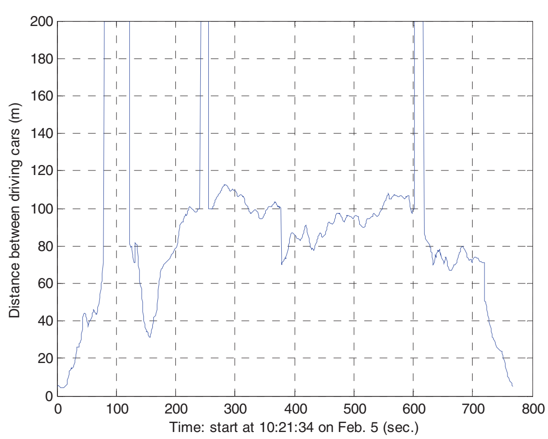

Figure 7 The distance between driving cars on February 5, 2010 start at AM10:21:34.

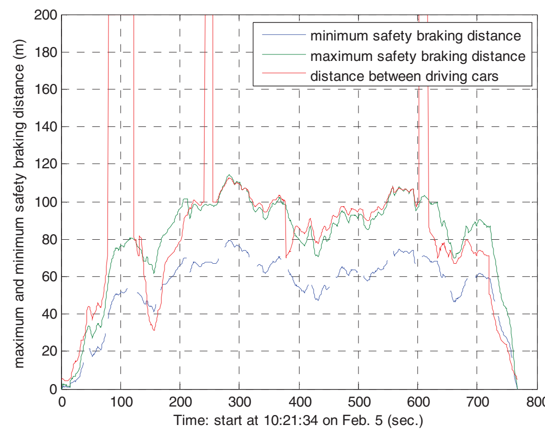

Figure 8 The maximum and minimum of safety braking distances of driving vehicles on February 5, 2010 start at AM10:21:34.

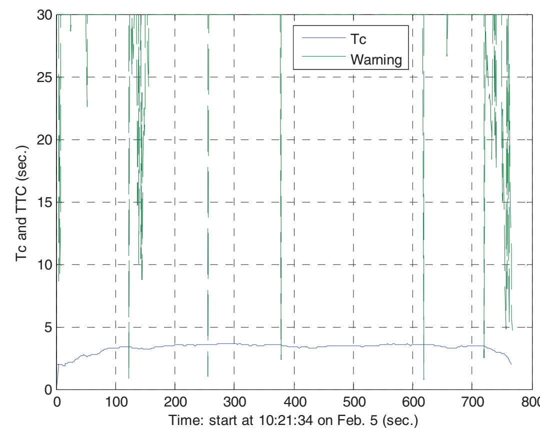

Figure 9 The TTC and critical TTC of vehicle operating on February 5, 2010 start at AM10:21:34.

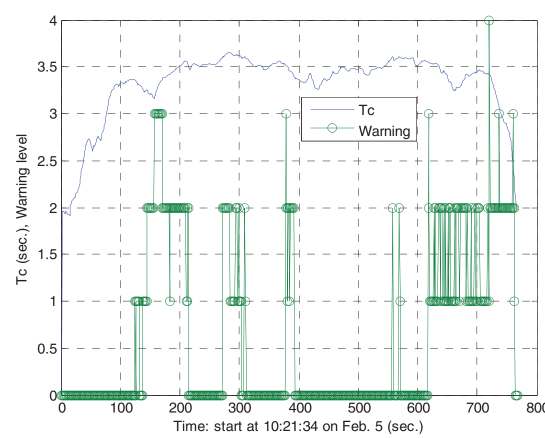

Figure 10 The critical TTC and warning signal of vehicle operating on February 5, 2010 start at AM10:21:34.

6. CONCLUSION

A novel algorithm considering both time- to-collision (TTC) and safety braking distance for forward collision warning system is presented for to alert and to assist driver keeping safety braking distance to avoid the collision accident in highway driving. The simulation results described the presented method is suitable applying in forward collision warning system for to avoid the forward collision accident. The useful alert messages can serve as a safety assistance system for safer driving in highway.

7. ACKNOWLEDGEMENT

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